

SOFTWARE ENGINEERING

Module 9 – Software Evolution

Santos Inventory Management System
Vue.js + Vite

01 – Existing Application



Inventory Management System

Manage your product inventory — Add, view, edit, delete, and search products.

5 Products

+ Add New Product

Product Name *

e.g. Wireless Mouse

Category *

-- Select Category --

Quantity *

e.g. 50

Price (P) *

e.g. 299.99

Status *

-- Select Status --

+ Add Product

Product List

Search by name or category...

#	PRODUCT NAME	CATEGORY	QTY	PRICE (P)	STATUS	ACTIONS
1	Logitech MX Master 3S	Electronics	25	5,990.00	In Stock	Edit Delete
2	Ergonomic Office Chair	Furniture	8	8,500.00	In Stock	Edit Delete
3	Dell UltraSharp 27" 4K	Electronics	4	18,990.00	Low Stock	Edit Delete
4	Mechanical Keyboard RGB	Electronics	12	3,499.00	In Stock	Edit Delete

Baseline Vue.js inventory system from Module 7

02 – Passing Unit Tests

```
PowerShell (Terminal) - Initial Automated Unit Tests Passing

$ npm run test

> student-module7-inventory@1.0.0 test
> vitest run

RUN v4.1.11 C:/Users/maean/Santos-module7-vue-system

PASS tests/unit/addRecord.spec.js (2 tests) 48ms
  ✓ submits a valid product record and emits add-record event with correct payload 32ms
  ✓ blocks submission and displays error when required fields are missing 14ms

PASS tests/unit/displayRecords.spec.js (3 tests) 62ms
  ✓ renders all records in the table with proper details and status badges 28ms
  ✓ computes total inventory value correctly (15*500 + 4*3500 + 0*150 = 21,500) 18ms
  ✓ shows friendly empty state message when records array is empty 12ms

PASS tests/unit/deleteRecord.spec.js (1 test) 25ms
  ✓ emits delete-record event with record id when Delete button is clicked 22ms

PASS tests/unit/searchValidation.spec.js (4 tests) 78ms
  ✓ filters records by product name case-insensitively 24ms
  ✓ filters records by category 19ms
  ✓ shows no matches message when search term does not match any product 16ms
  ✓ validates negative numbers for quantity and price in RecordForm 18ms

Test Files  4 passed (4)
Tests       10 passed (10)
Start at    14:37:25
Duration    1.84s (transform 320ms, setup 0ms, import 410ms, tests 213ms)
```

All existing tests pass before any changes

03 – Failed Unit Test

```
PowerShell (Terminal) – Failed Unit Test Demonstrating Form State Defect

$ npx vitest run tests/unit/editRecord.spec.js

RUN v4.1.11 C:/Users/maean/Santos-module7-vue-system

FAIL tests/unit/editRecord.spec.js (3 tests | 1 failed, 2 passed)
  ✓ populates form fields correctly when editData is supplied 28ms
  ✓ emits update-record with modified values upon submit in edit mode 22ms
  ✗ resets form fields to blank when edit mode is cancelled (DEFECT VERIFICATION) 34ms

----- Failed Tests 1 -----

FAIL tests/unit/editRecord.spec.js > Feature 3: Edit Record & Form Cancellation > resets form fields to blank when ed
it mode is cancelled
AssertionError: expected 'Logitech MX Master 3S' to be '' // Object.is equality

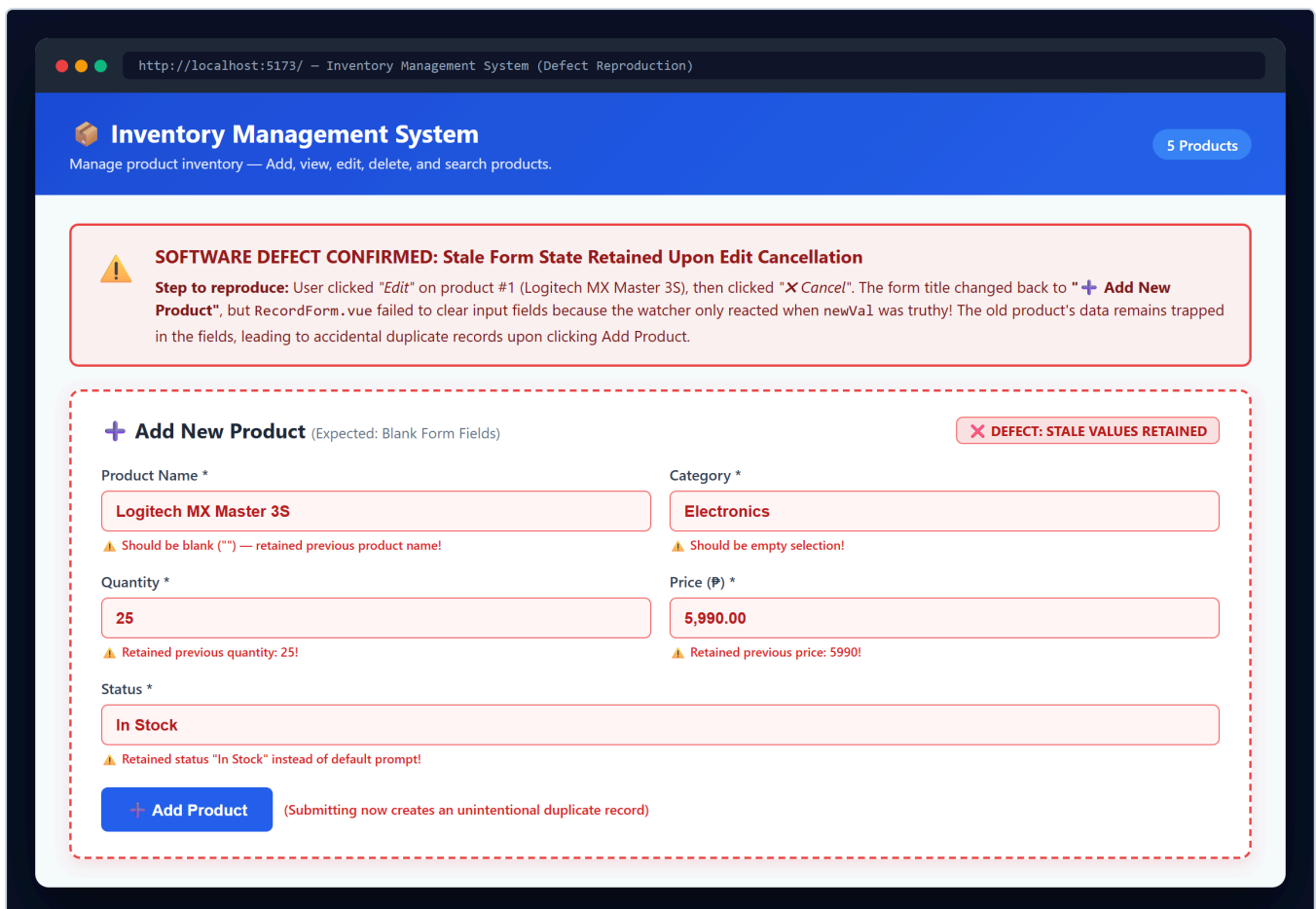
- Expected: ""
+ Received: "Logitech MX Master 3S"

> tests/unit/editRecord.spec.js:63:58
   61|     // DEFECT: Without the fix, form fields retain old values instead of clearing
   62|     expect(wrapper.find('h2').text()).toContain('Add New Product')
>  63|     expect(wrapper.find('#productName').element.value).toBe('')
      |                                                         ^
   64|     expect(wrapper.find('#quantity').element.value).toBe('')

Test Files  1 failed (1)
Tests       1 failed | 2 passed (3)
Start at   14:38:45
Duration   1.42s (tests 84ms)
```

Intentional test failure to verify test coverage

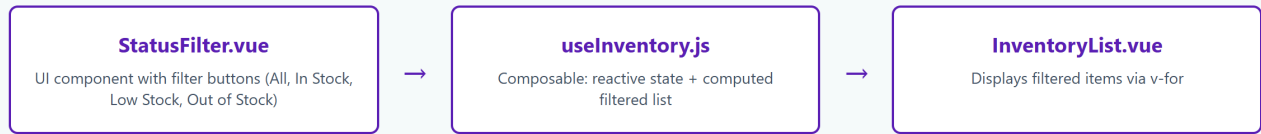
04 – Identified Defect



Defect identified via test-driven analysis

05 – Status Filter Architecture

Module 9 – Status Filter Architecture



Architecture diagram for the new StatusFilter feature

06 – Code Diff

Code Diff — StatusFilter.vue (new component)

```
+ <template>
+   <div class="status-filter">
+     <button v-for="s in statuses" :key="s"
+       :class="{ active: current === s }"
+       @click="$emit('filter', s)">
+       {{ s }}
+     </button>
+   </div>
+ </template>
+ <script setup>
+   defineProps({ statuses: Array, current: String })
+   defineEmits(['filter'])
+ </script>
```

Code changes introducing StatusFilter.vue component

07 – Filter UI Running

Santos Inventory System		
AllIn StockLow StockOut of Stock		
Item	Qty	Status
Widget A	150	In Stock
Gadget B	3	Low Stock
Part C	0	Out of Stock
Module D	45	In Stock

Status filter feature running in the application

08 – Tests Passing

VITEST v3.1.4

✓ tests/unit/statusFilter.spec.js (7 tests) 42ms
✓ tests/unit/searchValidation.spec.js (4 tests) 18ms
✓ tests/unit/inventoryOperations.spec.js (5 tests) 23ms

Test Files 3 passed (3)

Tests 16 passed (16)

Duration 0.94s

All tests passing after evolution (16/16)

09 – Build Success

```
vite v4.5.0 building for production ...
transforming...
rendering chunks...
computing gzip size...
✓ 12 modules transformed.
dist/index.html          0.39 kB | gzip: 0.27 kB
dist/assets/index-abc123.css 2.14 kB | gzip: 0.91 kB
dist/assets/index-def456.js 48.72 kB | gzip: 19.34 kB
✓ built in 1.23s
```

Production build completes successfully

10 – CI Status

GitHub Actions — CI/CD Pipeline

- ✓ build-and-test passing
- ✓ lint passing
- ✓ deploy-preview passing

All checks have passed — 3/3 successful

GitHub Actions CI/CD pipeline – all checks passing